Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/02/2018
PI Name:	Gonzalez, Adam Ph.D.		
Project Title:	Asynchronous Techniques for the Delivery of Empirically Supported Psychotherapies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	adam.gonzalez@stonybrook.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	347-564-5036
Organization Name:	State University of New York, Stony Brook		
PI Address 1:	Psychiatry, Stony Brook University-Putnam Hall		
PI Address 2:			
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-3365	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/01/2015	**End Date:**	06/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
Contact Email:	thomas.j.will1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2019 per NSSC information (Ed., 6/21/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:	August 2016: CoInvestigators Charles Boicey and Bireswar Loha have left Stony Brook Unviersity and are no longer working on the project. Their effort has been replaced by post-doctoral fellow (Minjung Shim, PhD).		
COI Name (Institution):	Kaufman, Arie Ph.D. (State University of New York, Stony Brook) Kotov, Roman Ph.D. (State University of New York, Stony Brook) Mahaffey, Brittain Ph.D. (State University of New York, Stony Brook)		
Grant/Contract No.:	NNX15AN64G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts are exposed to extreme environments and behavioral health risks during space missions. These stressors may place astronauts at elevated risk for behavioral health consequences including delirium, adjustment disorders, asthenia, and mood and anxiety problems, especially during long-duration missions. An expected barrier to delivering treatment to astronauts during long-duration missions is the lag in real-time communication between astronauts and mental health specialists on Earth. Therefore, the objective of this proposal is to identify technologies that will best facilitate the use of psychotherapeutic techniques when real-time communication is not possible. Phase I will involve systematic reviews of the existing literature on (a) psychotherapy techniques currently and historically used on space missions and; (b) existing technologies (e.g., video diaries, virtual reality immersion) with the potential for use in the asynchronous delivery of psychotherapeutic techniques to astronauts. Phase II will use the knowledge gained from Phase I to inform the design of a three-arm randomized clinical trial aimed at evaluating the feasibility, acceptability, and efficacy of the most promising technologies, alone and in combination, for asynchronous delivery of empirically supported psychotherapeutic skills. Products of this study will include best practice guidelines for asynchronous delivery of empirically supported psychotherapeutic techniques during long-duration space missions. These guidelines will be made public via a final report and peer-reviewed manuscript. Further, we will develop a presentation/workshop based on our findings to present to NASA behavioral health experts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will help inform best-practice guidelines for asynchronous communication for behavioral heath during long duration space missions.
Task Progress:	Goals and Objectives of the Reporting Period 1) Publish literature review in a peer-reviewed journal 2) Pilot the MyCompass program and trial procedures in preparation of a randomized clinical trial (RCT) 3) Recruit, enroll, and retain participants in RCT Progress Made During This Reporting Period 1) Publish literature review in a peer-reviewed journal. We have published the findings from our scoping review of RCTs examining the role of therapist support in Internet based Psychotherapeutic interventions for depression and anxiety disorders in Clinical Psychology Review. In addition, the process and findings of the review were presented at the Association for Behavioral and Cognitive Therapies conference in San Diego in November, 2017, and the NASA Human Research Program Investigators' Workshop in January, 2018. Shim, M., Mahaffey, B., Bleidistel, M., & Gonzalez, A. (2017). A scoping review of human-support factors in the context of Internet-based psychological interventions (IPIs) for depression and anxiety disorders. Clin Psychol Rev, 57, 129-140. doi:10.1016/j.cpr.2017.09.003 Shim, M., Mahaffey, B., Bleidistel, M., Hernandez, J., Gonzalez, A. (2017, November). A Review of Human-Support Factors in Internet-Based Behavioral Health Interventions for Depression and Anxiety Disorders. Association for Behavioral and Cognitive Therapies Annual Convention, San Diego, CA. Gonzalez, A., Mahaffey, B., Shim, M., Hernandez, J. (2018, January). Asynchronous Techniques for the Delivery of Empirically Supported Psychotherapies: Pilot Study. Poster presented at the NASA Human Research Program Investigators' Workshop, Galveston, TX. 2) Pilot the MyCompass program and trial procedures in preparation of a RCT. We pilot tested the study protocol with 9 participants to evaluate the feasibility and acceptability of the MyCompass program and the delayed communication platform. Participants' feedback was collected and any technological difficulties/challenges were resolved in preparation for the RCT. We collaboratively worked with the researchers at the Black dog Institute in Australia to optimize the web environment of the MyCompass program to be suitable to use by the participants of the RCT. We also worked with Stony Brook University Hospital IT to ensure all communications for the RCT were properly protected on a hospital-based secure server. 3) Recruit, enroll and retain participants in RCT. Based on the feedback and lessons learned from our pilot trial, we finalized the protocol and started recruiting participants for a 3-armed RCT (N=120) consisting of 3 asynchronous treatment delivery platforms: (1) self-management with MyCompass; (2) MyCompass + delayed therapist contact via email; and, (3) MyCompass + delayed therapist contact via video messaging. We also optimized our web portal to be used as a platform through which the study participants can complete weekly assessment and surveys as well as communicate (with at least a 44 minute delay) with their study clinician. The website has embedded Qualtrics surveys and allows both study participants and clinicians to upload text or video messages. We are actively recruiting "astronaut-like" participants experiencing elevated psychological distress from Stony Brook University campus and Stony Brook Hospital as well as major science laboratories in the Long Island area. To date, 135 people have been screened and 53 eligible participants have been randomly assigned to one of the three treatment conditions. The web-based MyCompass program will be upgraded to MyCompass 2.0 (improvements made to usability but no changes to treatment content) and released at the start of July 2018. We will shift to the upgraded program for the remaining participants. 4) Future Plans. We anticipate completing the RCT by October, 2018. Data analysis is ongoing and final analyses will be conducted after collection of the last follow up data. Dr. Adam Gonzalez, with support of the Co-Investigators, will be drafting the final report to NASA. In addition, Dr. Gonzalez will work with Dr. Raphael Rose to develop best practice guidelines for the asynchronous delivery of empirically supported psychotherapies on deep space missions. Deliverables will include the final report, the best practice guidelines, and peer-reviewed publication(s).
Bibliography Type:	Description: (Last Updated: 04/11/2022)

Abstracts for Journals and Proceedings	Shim M, Mahaffey B, Bleidistel M, Hernandez J, Gonzalez A. "A Review of Human-Support Factors in Internet-Based Behavioral Health Interventions for Depression and Anxiety Disorders." 51st Annual Convention of the Association for Behavioral and Cognitive Therapies, San Diego, CA, November 16-19, 2017. 51st Annual Convention of the Association for Behavioral and Cognitive Therapies, San Diego, CA, November 16-19, 2017. , Nov-2017
Abstracts for Journals and Proceedings	Gonzalez A, Mahaffey B, Shim M, Hernandez J. "Asynchronous Techniques for the Delivery of Empirically Supported Psychotherapies: Pilot Study." Poster presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Shim M, Mahaffey B, Bleidistel M, Gonzalez A. "A scoping review of human-support factors in the context of Internet-based psychological interventions (IPIs) for depression and anxiety disorders." Clinical Psychology Review. 2017 Nov;57:129-40. Epub 2017 Sep 12. Review. https://doi.org/10.1016/j.cpr.2017.09.003 ; PubMed PMID: 28934623 , Nov-2017